Supplement: hierarchical POMDP methods and parameters

This supplement makes the two-level construction of sec. 17.9 concrete: how location (L1) is coupled to context (L2) through context-conditioned priors,

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Supplement: hierarchical... (part 2)

what the alternating-minimization update actually computes, and the exact parameters of the executed run.

Supplement: hierarchical... (part 3)

It answers the question the main section brackets — *by what mechanism does a second latent level enter the inference at all* — and thereby fixes why the hierarchy resolves context above chance while retaining the separate location-accuracy contrast reported in sec. 17.9.

Generative model for context-gated location inference

The two-level POMDP is built by `build_hierarchical_world` in the `fedference.pomdp` module.

Generative model for... (part 2)

It couples the sentinel's 9-location L1 factor to a 2-state L2 context factor via **context-conditioned L1 priors**.

L1 (location). The standard 3x3 grid of `build_sentinel_world` has `n_s` = 9 states and sensor acuity 0.85.

Generative model for… (part 3)

L2 (context). A binary `quiet` / `alert` state has a symmetric transition matrix with persistence 0.90 and an initial uniform prior.

Generative model for... (part 4)

L1 priors given context. `quiet` is uniform over all 9 cells. `alert` places mass 0.60 at the center cell (the den) and spreads the residual uniformly.

Inference algorithm for top-down empirical priors

The module's `hierarchical_infer` function performs 4 passes of alternating minimization:

Inference algorithm for... (part 2)

1. L2 → L1 empirical prior.
$\tilde{\pi}_{0,\mathrm{L1}} = \sum_c q_{\mathrm{ctx}}[c]\, \pi_{0,\mathrm{L1}|c}$, a soft mixture of the two context-conditioned priors.

Inference algorithm for... (part 3)

2. L1 update. The one-step variational posterior is $q_{\text{loc}} = \mathrm{softmax}(\log \tilde{\pi}_{0,\mathrm{L1}} + \log A[\text{obs}, \cdot])$.

Inference algorithm for... (part 4)

3. L1 → L2 marginal evidence.
$\ell_c = \log(\pi_{0,\mathrm{L1}|c}^{\top} A[\mathrm{obs}, \cdot])$, the evidence for context c from the observed location likelihood.

Inference algorithm for... (part 5)

4. L2 update. $q_{\text{ctx}} = \text{softmax}(\log \pi_{0,\text{L2}} + \ell)$.

Inference algorithm for... (part 6)

After 4 iterations the agent broadcasts both q_{loc} and q_{ctx}; the colony federates each level independently via a log-linear pool (eq. 9).

Study parameters for the hierarchical condition

Parameter	Value
Agents	4
Trials	960
Acuity	0.85

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Study parameters for the... (part 2)

The executed hierarchical configuration is summarized in tbl. 17.